Multiplying polynomials

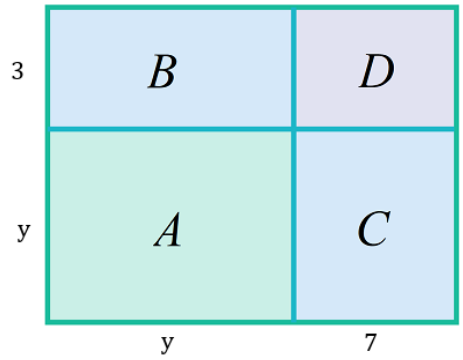
1 The area of the figure below is $(y+3)(y+7)$. We want to find any four smaller rectangles.

a Find the area of the following rectangles:

i Rectangle A **ii** Rectangle B

iii Rectangle C **iv** Rectangle D

b Using the areas of each rectangle, write an equivalent expression for the area of the figure.



2 Simplify the following:

a $(a+9)(b-6)$

b $(4r+7)(7s+2)$

c $(x+2)(x+5)$

d $(x+5)(x+7)$

e $(v+8)(v+10)$

f $(x+5)(x-2)$

g $(x-2)(x-3)$

h $(m-6)(m+3)$

i $(5m+8)(m-3)$

j $(7w+5)(5w+2)$

k $(7y-6)(6y+6)$

l $(0.5c-7)(8c-0.4)$

m $\left(\frac{1}{3}y+\frac{1}{4}\right)\left(\frac{1}{3}y-\frac{1}{6}\right)$

n $(9y-8)(-10y-6)$

o $9(y+8)(y+2)$

p $7(x-9)(x+3)$

q $-(x+5)(x+2)$

r $-5(y+4)(y+5)$

s $-4(y-7)(y+4)$

t $5(3x-2)(-5x+4)$

u $3(2x-6)(x+1)$

v $\left(2x^5+y^4\right)\left(3x^5+5y^4\right)$

w $6(-5x+3)(x+2)-2$

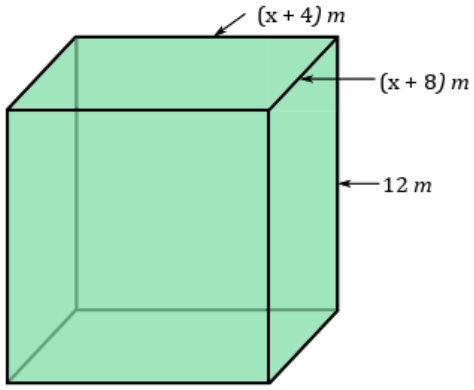
x $2(-4x-6)(x-6)+4x$

y $3\left(6x+\frac{1}{2}\right)(x-2)-5$

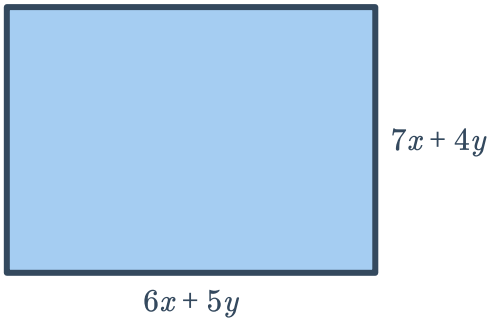
z $-8\left(\frac{x}{2}+2\right)(x-2)+2x$

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3 Consider the rectangular prism shown with its dimensions labeled.
Write a fully simplified expression for the volume of the prism.

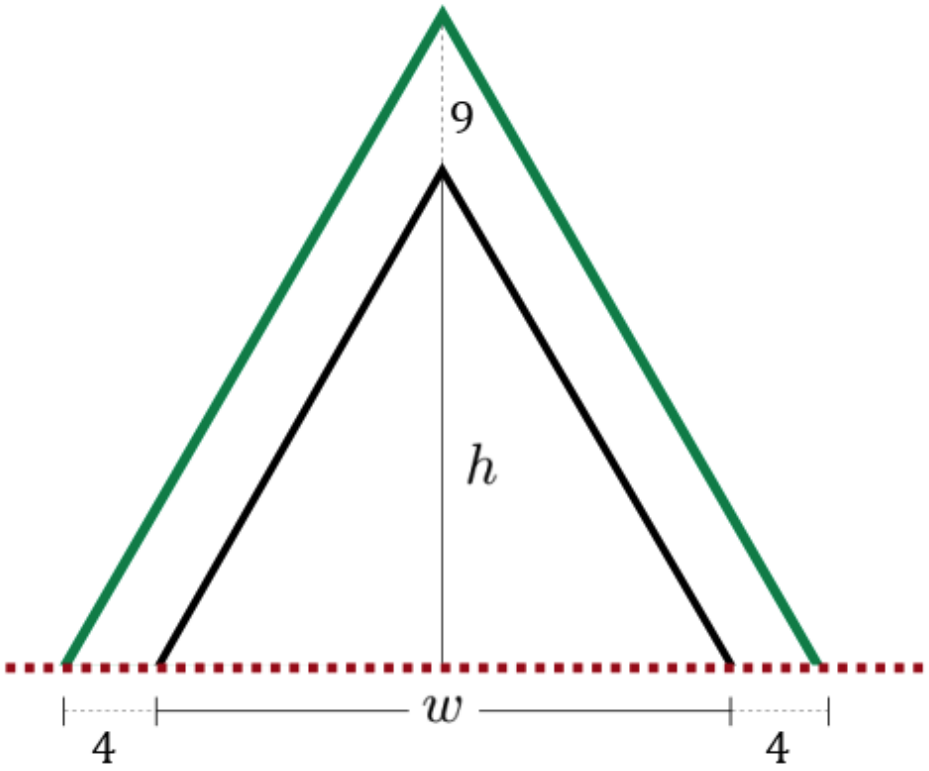


4 Write a polynomial in two variables that describes the total area of region shaded in blue.



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5 Elizabeth wants to know the difference in cross-sectional area of the door of her tent and the tent itself. The proportions of the tent are exactly the same as the door, as seen in the picture below. The triangular door has a length and height of w and h respectively.
The top of the tent is 9 cm higher than the door, and its ends are 4 cm longer than the ends of the door.



- **a** Express the height of the tent in terms of h.
- **b** Express the length of the tent in terms of w.
- **c** Hence, find a simplified expression for the difference between the cross-sectional area of the tent and the tent door.

6 Han is trying to determine the dimensions of a rectangular box that will result in the largest volume. The height of the box is 4 cm, the width is $4y$ cm and the length is 3 cm more than double the width.
Write a fully simplified expression for the volume of the box.

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7 Simplify the following:

a $(a+2)\left(5a^2-2a+2\right)$

b $(u-4)\left(5u+5+u^2\right)$

c $(5a+5)\left(-5a-4-5a^2\right)$

d $\left(5c^2-2\right)\left(4c^2+2-3c\right)$

e $(2a-3b)\left(3a^2-2ab-4b^2\right)$

f $\left(-3m^2-2m+2\right)(4m+4)$

g $(-7x+3y+9)(8x+12)$

8 Consider the statement: $(u+1)\left(u^2-u+1\right)=u^3+1$
Is this statement true or false?

9 Without doing the multiplication completely, find the coefficient of x in

$$\left(x^2-3x-5\right)(5x-4)$$

10 Consider $R(x)$, the product of the polynomials $P(x)=3x^5-3$ and $Q(x)=-2x^7+5x^5+6$.

a Find the degree of $R(x)$.

b Find the constant term of $R(x)$.

c Is $R(x)$ a polynomial?

11 When $P(x)=3x^4$ is multiplied by $Q(x)$, the resulting polynomial is $P(x)\cdot Q(x)=15x^8$.

a Find the degree of the polynomial $Q(x)$.

b Find the leading coefficient of the polynomial $Q(x)$.

12 Given that $P(t)=3-4t^4+5t^2$ and $Q(t)=4t^2-3t^3$.
Find the leading term of $P(t)\cdot Q(t)$.

13 $P(x)$ and $Q(x)$ are two polynomials.
Determine whether the following could be the result of $P(x)\cdot Q(x)$.

a $5x^4-3x^6+6$

b $5x^{\frac{1}{4}}+6x^2+4$

c $3x^4+\dfrac{4}{x^9}-\dfrac{7}{6}$

d $\dfrac{1}{x^2+3x-3}$

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14 Both $g(x)$ and $h(x)$ are polynomial functions.
Determine whether the following expressions must also be polynomials.

a $\dfrac{1}{h(x)} + \dfrac{1}{g(x)}$

b $h(x)(g(x) - h(x))$

c $g(x) + h(x)$

d $\dfrac{h(x)\,g(x)}{h(x) + g(x)}$

e $\dfrac{h(x) - g(x)}{h(x)} + \dfrac{h(x) + g(x)}{h(x)}$

f $\dfrac{g(x) - h(x)}{g(x)}$

15 Determine whether the following statements are true when $(x)(x-3)(2x+6)$ is expressed as a polynomial in standard form.

a The leading coefficient is 2

b The degree is 2

c The number of terms is 2

d The constant term is 2

16 Determine whether the following are equivalent to $2x^2 + 12x + 16$.

a $(2x+2)(x+8)$

b $2(x+4)(x+2)$

c $(2x+4)(x+4)$

d $(2x+8)(x+2)$

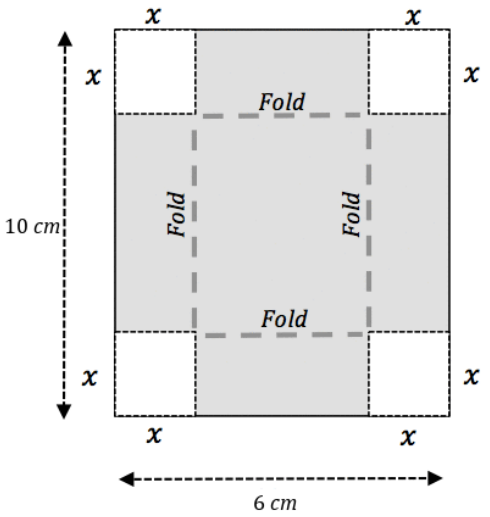
17 A landscape design company makes large raised boxes for plantings. The height of the boxes they build is $(2y+3)$ ft, and the area of the rectangular base measures $(4y^2 + 5y + 1)$ ft^2.
If the box is filled with soil, find the volume of soil needed.

18 Consider a rectangular cardboard measuring 10 cm by 6 cm. It is to be converted into a box with no lid by cutting out square corners measuring x cm in length and folding up the sides.
Find the expression for the following in terms of x. Leave your answer in expanded form.

a The length of the cardboard box.

b The width of the cardboard box.

c The volume of the cardboard box.



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Additional questions

19 Find the missing values that make each equation true.

a $(x+\square)(x-7)=x^2-3x+\square$

b $(2x-1)(\square x-5)=2x^2+\square x+5$

c $(\square x+3)(\square x+6)=12x^2+30x+\square$

20 Sonya wants to frame a 5×7 picture frame. If the frame is meant to have a constant width of x all around, write an expression for

a The new dimensions of the picture with its frame.

b The area of the picture with its frame.

c The perimeter of the picture with its frame.

21 A flat rate large box from USPS has dimensions of $12\times 12\times 5.5$ inches. If Jerome decides to put a layer of insulation in his box x inches thick, write a simplified polynomial expression that models

a Each dimension of the open volume left in the box.

b The volume he has left in his box to fill.

22 Anette decides to use the area model to complete a polynomial multiplication question, $(10x+2)(10x+3)$. To warm-up, she decides to try a simple integer example:

$12(13)=$

	10	2
10	100	20
3	30	6

$100+20+30+6=156$

$(10x+2)(10x+3)$

	$10x$	2
$10x$	$100x$	$20x$
3	$30x$	6

$100x+20x+30x+6=150x+6$

a What is the error in Anette's polynomial product? Correct her work and solution.

b How would you set up the area model to find the product $(10x+2)(x^2+10x+3)$?

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c Find the product using any method.

23 If Mackenzie creates a polynomial with degree 4 and Edgardo creates a polynomial with degree 3, what do we know about the product of their polynomials?

24 Consider the following problem: "Find two binomials whose product results in a polynomial with 5 terms".

a Yao was given the problem, but they think it's impossible. Do you agree or disagree? Explain.

b Create two different products of two polynomials that results in an answer with exactly 5 terms.

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